

Calculus 2 - Homework 5 (Problems)

1.

Evaluate $\int \frac{1}{(x-1)(x+1)} dx$.

2.

Evaluate $\int \frac{1}{x^2 + x} dx$.

3.

Determine whether $\int_2^{\infty} \frac{1}{x (\ln x)^2} dx$ converges, and if so find its value.

4.

Evaluate $\int \cos^3(x) dx$.

5.

Evaluate $\int x e^x dx$.

6.

Determine whether $\sum_{n=1}^{\infty} \frac{\sin^2(n)}{n^2}$ converges or diverges.

7.

Evaluate $\int \frac{(3x+1)}{(x+1)(x^2+1)} dx$.

8.

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{2^n}{n^2}$ converges or diverges.

9.

Use the Trapezoidal Rule with $n = 4$ subintervals to approximate $\int_0^2 x^2 dx$. Report the estimate.

10.

Evaluate $\int \frac{x}{x^2 + 4} dx$.